

9. With A and p as in Exercise 7, determine if p is in $\text{Nul } A$.

Solution: By Defn, $\text{Nul } A$ refers to the set of vectors x s.t. $Ax=0$.

$$Ap = \begin{bmatrix} 2 & -3 & -4 \\ -8 & 8 & 6 \\ 6 & -7 & -7 \end{bmatrix} \begin{bmatrix} 6 \\ -10 \\ 11 \end{bmatrix} = \begin{bmatrix} 2 \cdot 6 + (-3) \cdot (-10) + (-4) \cdot 11 \\ -8 \cdot 6 + 8 \cdot (-10) + 6 \cdot 11 \\ 6 \cdot 6 + (-7) \cdot (-10) + (-7) \cdot 11 \end{bmatrix} = \begin{bmatrix} -2 \\ -62 \\ 29 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore p \notin \text{Nul } A. \quad \square$

10. In Exercises 11 and 12, give integers p and q such that $\text{Nul } A$ is a subspace of $\mathbb{R}^p$ and $\text{Col } A$ is a subspace of $\mathbb{R}^q$. 3×4

$$11. A = \begin{bmatrix} 3 & 2 & 1 & -5 \\ -9 & -4 & 1 & 7 \\ 9 & 2 & -5 & 1 \end{bmatrix}$$

Solution: A is a ~~3x3~~ matrix.

By Defn, $\text{Col } A$ refers to the set of all linear combinations of the columns of A . The columns of A are in $\mathbb{R}^3$.

$$\therefore q=3.$$

By Defn, $\text{Nul } A$ refers to the set of all solutions of the matrix equation $Ax=0$. By Defn. of Ax , $\because A$ is a 3×4 matrix, x must be a 4×1 vector.

$$\therefore p=4 \quad \square$$

13. For A as in Exercise 11, find a nonzero vector in $\text{Nul } A$ and a nonzero vector in $\text{Col } A$.

Solution:

$$\begin{aligned} & 1 \begin{bmatrix} 3 \\ -9 \\ 9 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 7 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix} \in \text{Col } A. \end{aligned}$$

We need to solve the matrix equation $Ax=0$. By Theorem 3 from Chapter 1, this has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.